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Department of Electronics and Communication Engineering

A Major Project Abstract Review on

Design and Verification of AXI/APB Bus Protocol using UVM

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Introduction

- Modern SoCs integrate multiple IP blocks (CPU, memory, peripherals) using ARM's AMBA protocol family.
- These components communicate using the AMBA bus protocols developed by ARM.
- AXI (Advanced eXtensible Interface) – high-performance bus for CPU, memory, DMA.
- APB (Advanced Peripheral Bus) – low-power bus for simple peripherals (UART, timer, GPIO)
- Since AXI and APB differ in timing/handshake, a bridge is required to connect them
- Verification of this bridge is critical, and UVM (Universal Verification Methodology) is used since it's the industry-standard reusable framework for such verification

Objective

- To design an AXI-to-APB bridge at RTL level using Verilog/SystemVerilog.
- To develop a complete UVM-based verification environment for the bridge.
- To generate constrained-random and directed test cases covering normal, burst, and error scenarios.
- To build a self-checking scoreboard for automatic result comparison.
- To achieve maximum functional and code coverage.
- To ensure protocol compliance and reliable communication.
- To convert AXI transactions into APB transactions.

Literature Survey

- Existing research focuses on the design and verification of AMBA bus protocols for SoC communication.
- Most works use AXI for high-speed communication and APB for low-power peripheral communication.
- Recent studies use UVM (Universal Verification Methodology) to automate the verification process.
- UVM-based verification improves test reusability, debugging, and functional coverage.
- Verification techniques such as constrained-random testing, scoreboard, assertions, and coverage analysis are widely used.
- The literature shows that UVM significantly reduces verification time compared to traditional testbenches.

Existing System

- Uses conventional Verilog/VHDL directed testbenches for verification.
- Test cases are written manually, making verification time-consuming.
- No constrained-random testing, so corner cases may be missed.
- Manual waveform checking makes debugging slow and error-prone.
- Verification components are not reusable for different projects.
- Provides limited functional and code coverage.
- Results in higher verification effort, longer development time, and lower verification efficiency.

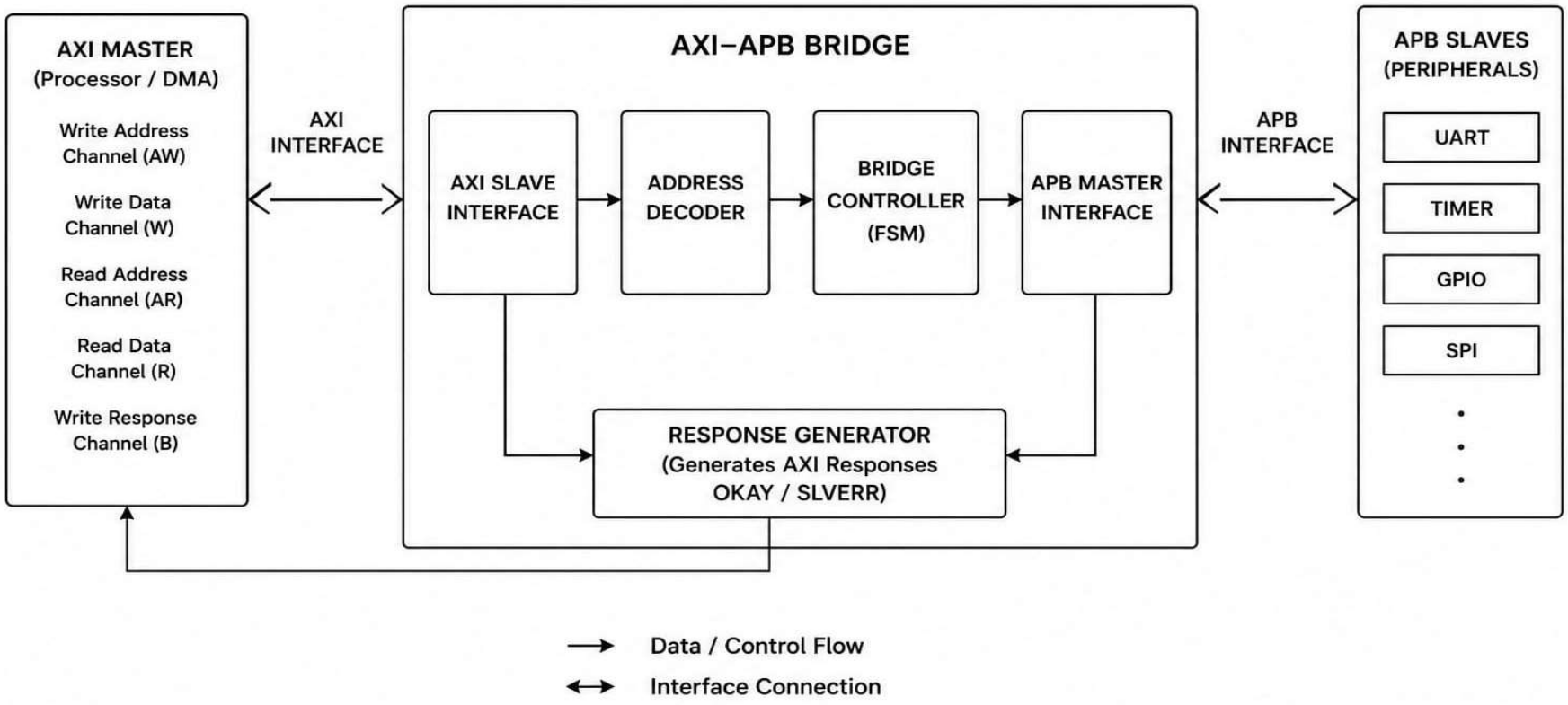
Proposed System

- Design an AXI-to-APB Bridge using Verilog/SystemVerilog.
- Develop a UVM-based reusable verification environment.
- Generate constrained-random AXI transactions automatically.
- Convert AXI transactions into APB transactions through the bridge.
- Monitor protocol behavior using UVM monitors.
- Compare expected and actual outputs using a scoreboard.
- Measure verification completeness using functional and code coverage.

Software Specifications

- Design Language: Verilog / SystemVerilog
- Verification Methodology: UVM 1.2 (SystemVerilog Class Library, IEEE 1800.2 standard)
- Simulator: Cadence Xcelium / Synopsys VCS / Mentor QuestaSim
- Waveform Debug Tool: SimVision / Verdi
- Coverage Tool: IMC (Integrated Metrics Center) / Verdi Coverage
- Operating System: Linux (Ubuntu/CentOS)
- Scripting (optional): Makefile / Shell / Perl for regression automation

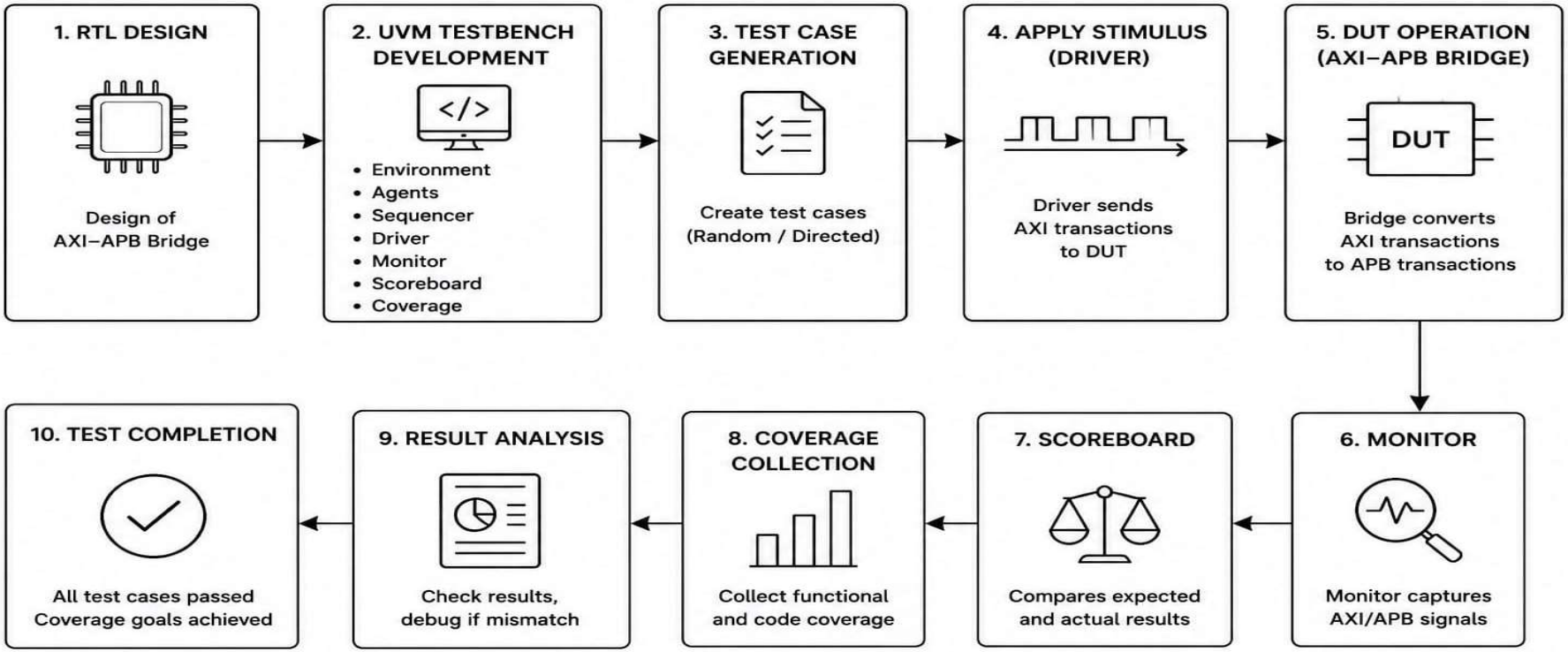
Architecture of AXI-APB Bridge



Working Methodology

- AXI Master initiates read/write transactions.
- AXI Slave Interface receives AXI requests and handles communication.
- Address Decoder identifies the target APB peripheral.
- Bridge Controller (FSM) converts AXI transactions into APB transactions.
- APB Master Interface transfers data to the selected APB peripheral.
- APB Slaves (UART, Timer, GPIO, SPI) perform the requested operation.
- Response Generator sends OKAY or SLVERR status back to the AXI Master

UVM Verification Flow Diagram



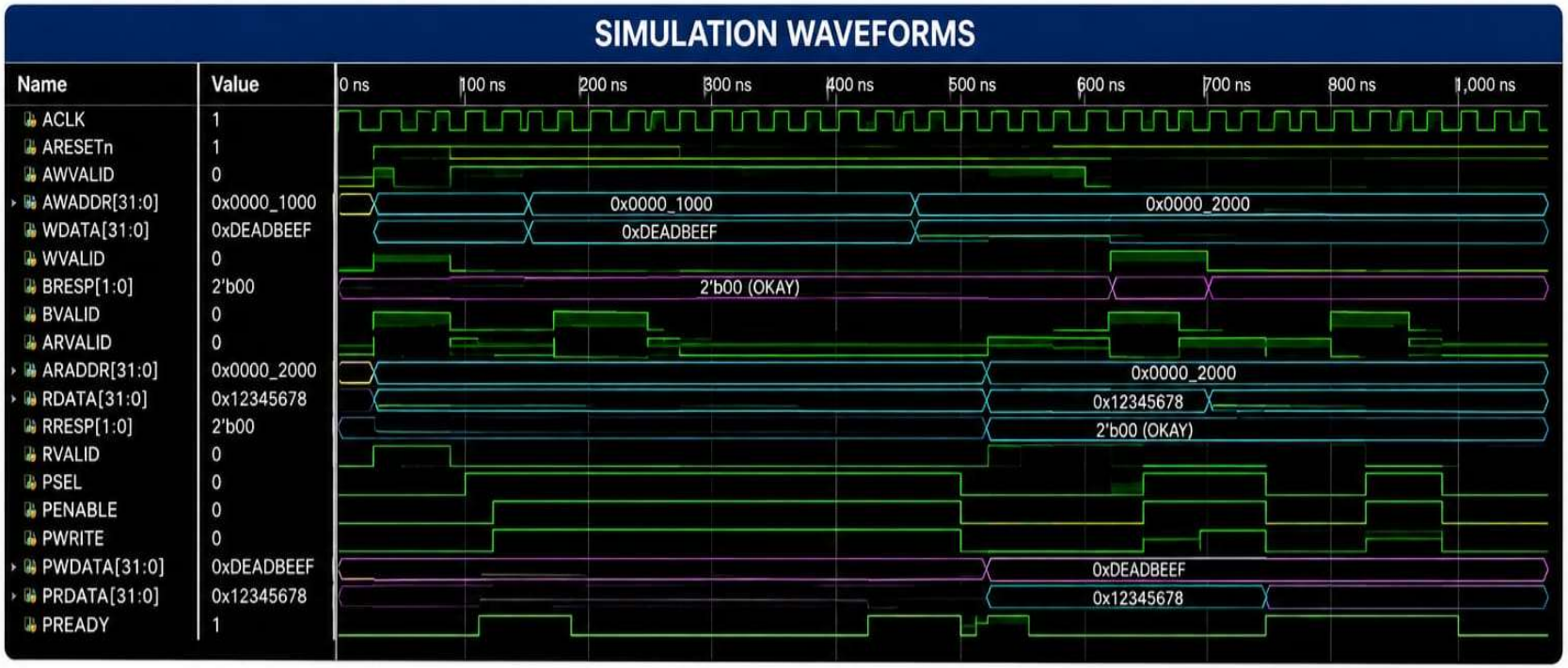
Working Methodology

- RTL Design – Design the AXI–APB Bridge using Verilog/SystemVerilog.
- UVM Testbench Development – Create the UVM environment with driver, monitor, scoreboard, and coverage components.
- Test Case Generation – Generate directed and constrained-random test cases.
- Apply Stimulus (Driver) – Driver sends AXI transactions to the Design Under Test (DUT).
- DUT Operation – The AXI–APB Bridge converts AXI transactions into APB transactions.
- Monitor & Checking – Monitor captures AXI/APB signals and forwards them for verification.
- Scoreboard – Compares expected results with actual DUT outputs.
- Coverage Collection – Measures functional and code coverage to ensure complete verification.
- Result Analysis – Analyze simulation results and debug any mismatches.
- Test Completion – Verification is completed after all test cases pass and coverage goals are achieved

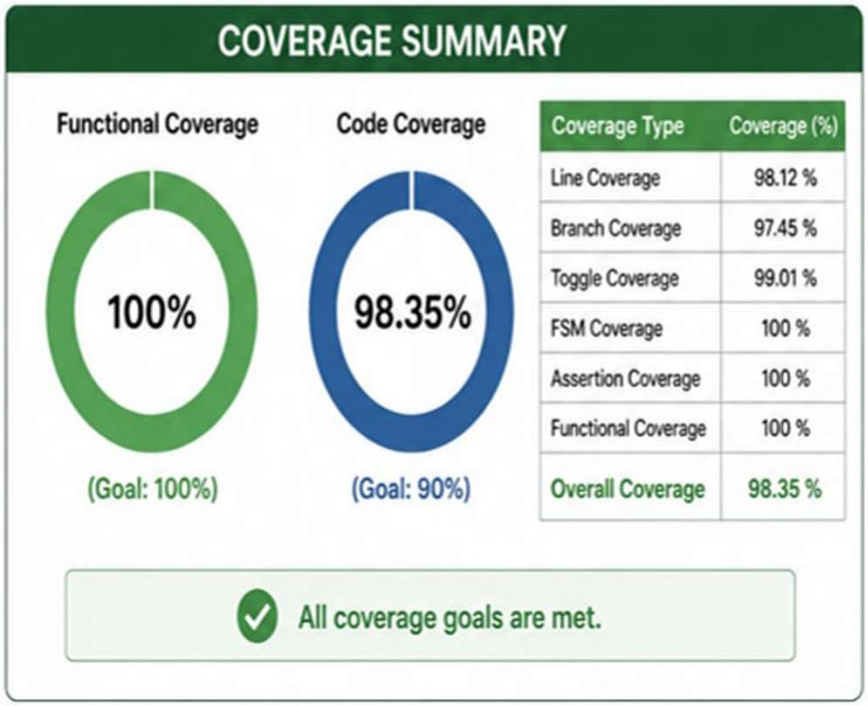
Conclusion

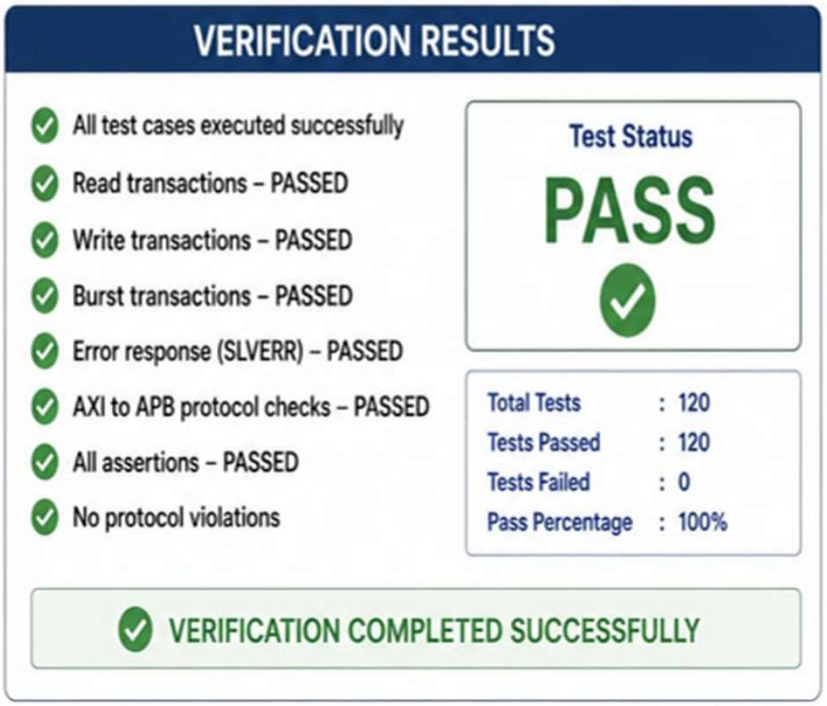
- Designed an AXI–APB Bridge for communication between high-speed AXI and low-speed APB buses.
- Developed a UVM-based verification environment to verify the bridge functionality.
- Verified read, write, burst, and error transactions using automated test cases.
- Improved verification efficiency through reusable components and constrained-random testing.
- Achieved high functional and code coverage, ensuring reliable and error-free operation.
- Demonstrates the importance of UVM in modern SoC verification by reducing verification time and improving design quality.

Expected Waveforms



Coverage Summary and Verification Results





Future Scope

- Extend the design to support AXI4 Full protocol and multiple APB peripherals.
- Implement the design on an FPGA for real-time hardware validation.
- Integrate Assertion-Based Verification (SVA) to improve error detection.
- Enhance the UVM environment with advanced constrained-random and regression testing.
- Reuse the developed Verification IP (VIP) for other AMBA-based protocols such as AHB and AXI-Lite.
- Apply the verification methodology to complex SoC designs for improved reliability and faster verification.

References

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